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SC388: AL and ML Interpretability

May 18, 2026

I. INTRODUCTION

Accurately predicting individual summit success in high-altitude Himalayan mountaineering is of substantial interest to climbers, expedition organizers, guides, and medical support teams. Misjudging the likelihood of success can result in considerable financial losses for commercial expeditions or even serious safety risks. To investigate this problem, we analyze the Himalayan Climbing Expeditions (HimEx) database and model the dataset of individual climber attempts. Using these data, we fit two complementary Random Forest models. The first is a binary classification model predicting summit success, and the second is a regression model fit exclusively to failed attempts, predicting the proportion of the mountain climbed. Both global and local interpretability methods, as well as uncertainty quantification were used to provide insight into the factors driving the model's predictions using multiple levels of model confidence.

II. METHODS

Three source tables were merged to construct the modelling dataset. Individual member records were left-joined to peak metadata on peak_id to retrieve peak height in metres, then left-joined to expedition logs on expedition_id to retrieve expedition year. From the merged dataset, nine variables were retained: the binary outcome success, climber demographics (age, sex), logistical indicators (oxygen_used, hired, solo), environmental context (season,

height_metres), and the temporal covariate year. Then, rows containing missing or ambiguous values were removed. The six categorical variables (success, sex, season, oxygen_used, hired, and solo) were also converted to factors prior to modelling.

The cleaned dataset contained 72,014 individual climber records with an overall success rate of approximately 38% (Appendix 5.8). The data were randomly partitioned into an 80% training set and a 20% hold-out set. The hold-out set was later divided equally into calibration and final evaluation partitions for the conformal prediction analysis. A limitation of this partitioning strategy is that climbers and expeditions from different historical periods may appear across both training and evaluation sets. Consequently, reported predictive performance should be interpreted as measuring interpolation across the observed data distribution rather than forecasting future expedition outcomes under strict temporal separation.

A Random Forest classification model was fit to the training data using 500 trees, with success as the binary outcome and the remaining eight variables as predictors. To partially address class imbalance, inverse class-frequency weights were computed by assigning higher weight to the minority class, and passed to the classwt argument of randomForest. Variable importance was recorded with importance = TRUE and is reported via both mean decrease in accuracy and mean decrease in Gini (Appendix 5.7). Predictive performance was first estimated through 5-fold stratified cross-validation on the training set, implemented using createFolds from the caret package with the same class weights applied within each fold, yielding a mean cross-validated accuracy of 75.8% ± 0.25% SD.

Global SHAP values for the classification model were computed over the full test set using fastshap::explain with 50 simulations and bias adjustment enabled (adjust = TRUE). The

prediction wrapper extracted the model's estimated probability of summit success for each observation. Results were visualized using shapviz as a beeswarm summary plot ranking all eight features by their SHAP contribution and showing directional effects through a low-to-high feature value colour gradient (Appendix 5.6). Two dependence plots were additionally produced to examine interaction effects between the most contextually meaningful feature pairs: peak height coloured by oxygen use (Fig 3), and age coloured by hired status (Fig 2).

Split conformal prediction was applied to the classification model at a nominal miscoverage level of $\alpha = 0.10$. The hold-out set was divided in half at random to produce a calibration partition and a final evaluation partition. Non-conformity scores were computed on the calibration partition as $1 - \hat{p}$ for true positive cases and $\hat{p}$ for true negative cases. The 90th percentile of the remaining calibration scores was retained as the threshold $\hat{q}$. On the evaluation partition, the TRUE class was included in a prediction set if $(1 - \hat{p}) \leq \hat{q}$, and the FALSE class if $\hat{p} \leq \hat{q}$. Empirical coverage on the evaluation set was 89.47%, marginally below the nominal 90% guarantee.

SHAP waterfall plots were generated for seven cases selected from the evaluation partition (Appendix 5.3.1–5.3.7). Uncertainty was defined as observations with a predicted probability within 0.10 of 0.5. The three most uncertain cases (sorted by proximity of $\hat{p}$ to 0.5), the two highest-confidence predicted successes (sorted by descending $\hat{p}$ among correctly classified true positives outside the uncertainty band), and the two highest-confidence predicted failures (sorted by ascending $\hat{p}$ among correctly classified true negatives outside the uncertainty band) were selected. SHAP values for these seven observations were computed separately using fastshap::explain with 20 Monte Carlo simulations on the evaluation features, and individual waterfall plots were produced via sv_waterfall.

A secondary analysis was conducted on failed attempts only. Records where success was FALSE were retained from the merged dataset, and cases missing highpoint_metres or height_metres, or where height_metres was zero, were excluded. The continuous outcome percent_climbed was defined as highpoint_metres / height_metres, capped at 1 via pmin. The distribution of this outcome among failures is shown in Appendix 5.2. The same eight predictors used in the classification model were retained, with identical factor coercions and cleaning filters applied. The failure dataset was split 80/20 into training and test sets, and a 500-tree Random Forest regression model was fit to the training data with importance = TRUE and no class weighting. SHAP values were computed on the test features using fastshap::explain with 50 simulations and bias adjustment enabled, and visualized as a beeswarm importance plot ranking the predictors of how far unsuccessful climbers progressed before turning back (Appendix 5.1).

III. RESULTS

The analysis has two components. The first is a Random Forest classification model predicting summit success. The second is a Random Forest regression model fit exclusively to failed attempts, predicting how far a climber progressed before turning back. Global and local interpretability were applied to the classification model through SHAP. Uncertainty was quantified through split conformal prediction. The Random Forest achieved 75.8% mean 5-fold cross-validated accuracy (SD = 0.25%) against a 62% baseline from always predicting failure. Conformal prediction at $\alpha = 0.10$ produced 89.47% empirical coverage, just under the nominal 90% guarantee. 2,796 of 7,302 evaluation observations (38.3%) were assigned both TRUE and FALSE in the prediction set, meaning the model withheld a confident prediction for those climbers.

SHAP values ranked oxygen use as the dominant predictor, with contributions for oxygen_used (TRUE) reaching +0.55, larger than any other feature by a wide margin (Appendix 5.6). Hired status ranked second at +0.10 to +0.25.

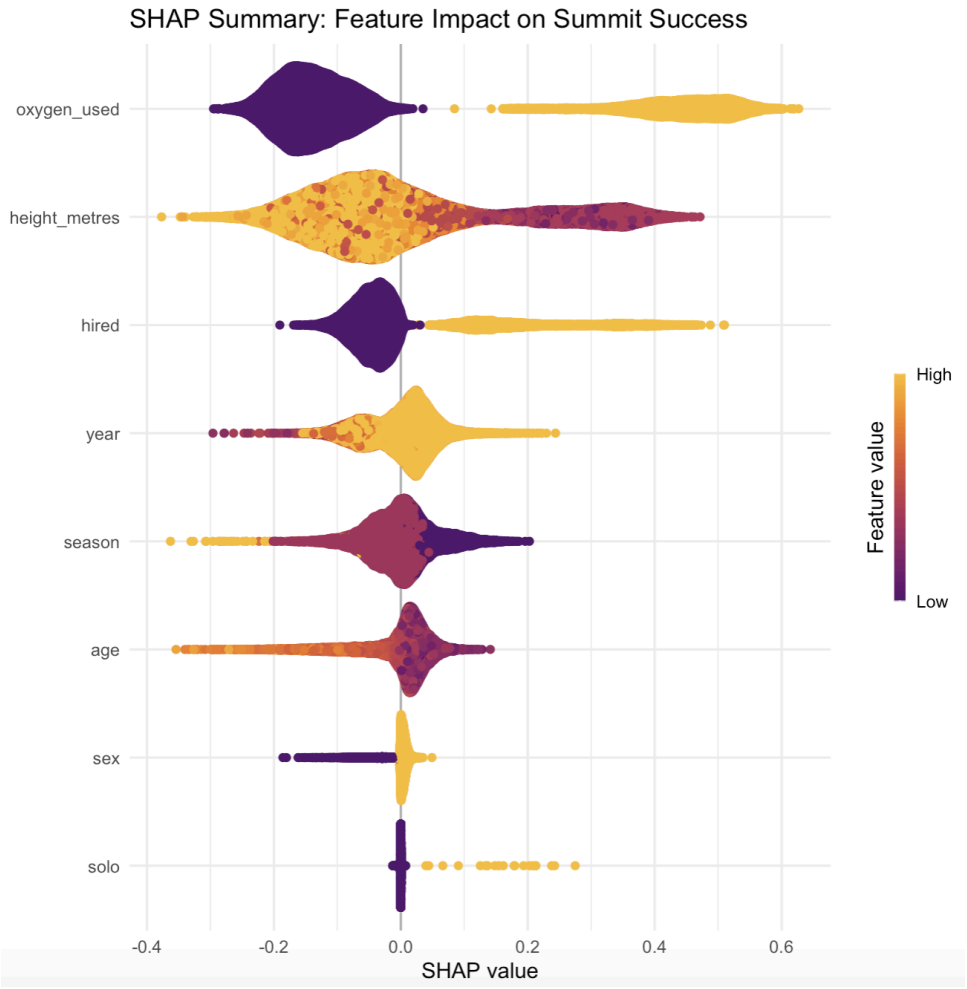


Fig: 1: *SHAP summary plot ranking all eight features by their contribution to predicted summit success probability. Each point is one climber in the test set. Oxygen use has the widest spread, with TRUE values (yellow) pushing predictions up to +0.55 and FALSE values (purple) pulling down to -0.40. Hired status and peak height follow, with hired TRUE consistently positive and higher elevations consistently negative. Age, season, and year have narrower distributions centered near zero. Sex and solo have the smallest effects across the test set.*

Peak height penalized predictions above 8,000 m, but the height-by-oxygen dependence plot shows oxygen substantially offsets this penalty at a threshold rather than linearly (Figure 3).

Age produced negative contributions past 45, compounded for climbers without guides. The age-by-hired dependence plot shows the guide effect is near zero below 35 and grows to 0.10 to 0.15 SHAP units by age 60 (Fig 2). Year, season, sex, and solo status had smaller but directionally consistent effects.

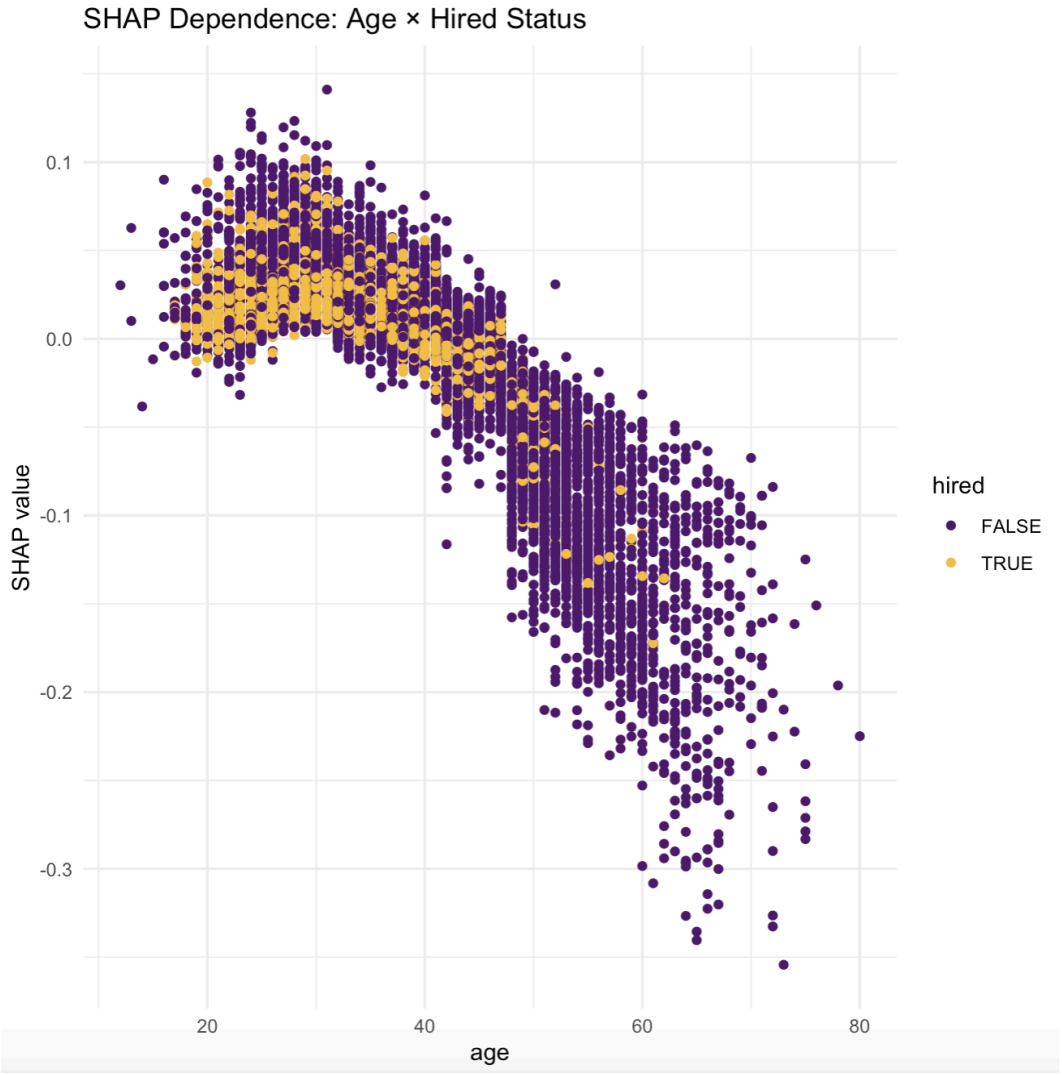


Fig 2: *SHAP dependence plot of age and hired status. The gap between hired (yellow) and non-hired (purple) is near zero below 35 and grows to 0.10 to 0.15 SHAP units by age 60. Older climbers without guides show the steepest decline in predicted success.*

Waterfall plots for seven selected observations decomposed individual predictions into feature contributions (Appendix 5.3.1–5.3.7). The selection covered the three most uncertain cases near $\hat{p} = 0.5$, the two highest-confidence predicted successes, and the two

highest-confidence predicted failures. High-confidence failures were dominated by extreme altitude without oxygen. Furthermore, high-confidence successes showed large positive contributions from oxygen and hired status together, and the uncertain cases had conflicting contributions with no dominant signal, consistent with the conformal prediction sets assigning both outcomes to those observations.

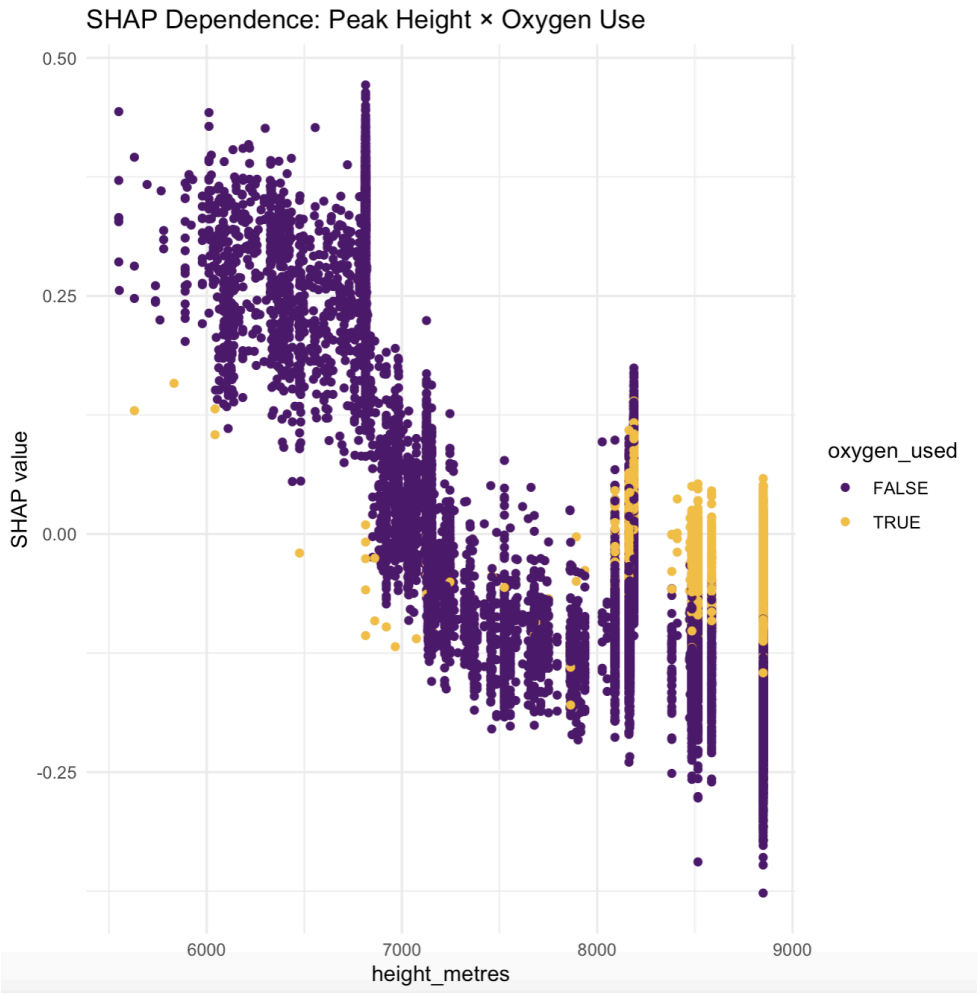


Fig 3: *SHAP dependence plot of peak height and oxygen use. Purple points are climbers without oxygen, yellow with oxygen. The gap between the two opens at ~8,000 m and widens toward 9,000 m.*

A secondary model was fit on failed attempts only to understand where climbers fail instead of just focusing on if they fail. The regression model on failed attempts identified oxygen use and peak height as the strongest predictors of how far a climber progressed before turning

back. The percent_climbed distribution is right-skewed, with most failures above 85% of peak height. Observations at 100% are coded as failures because the climber did not survive the descent.

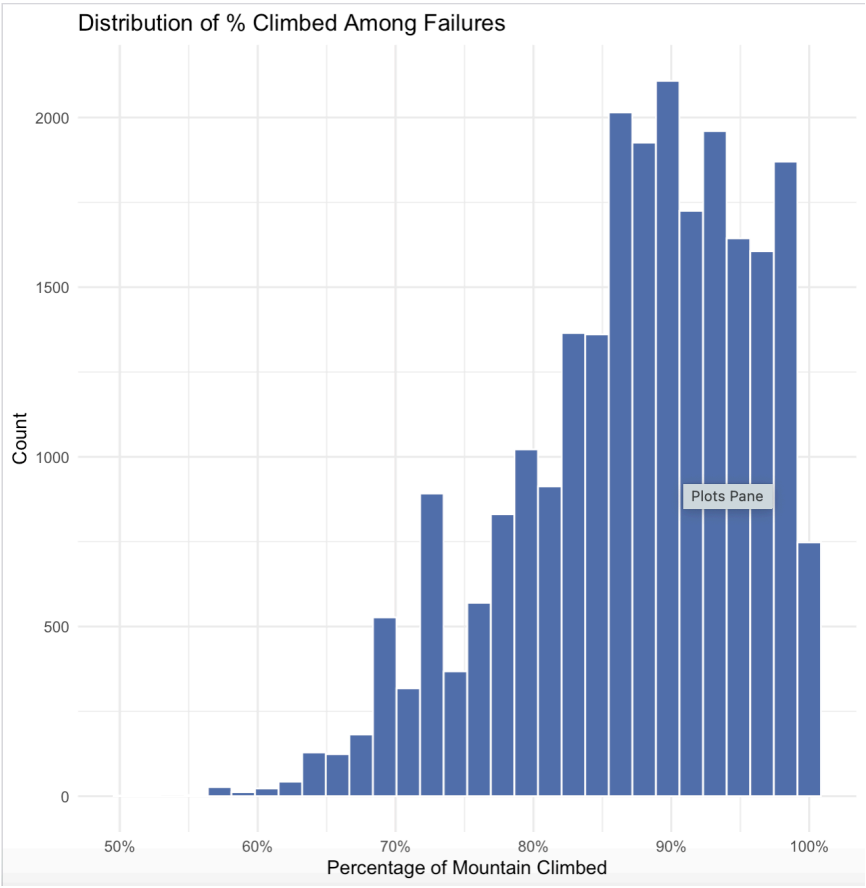


Fig 4: *Distribution of percent climbed among failed attempts. Most failures occur above 85% of peak height. Observations at 100% are coded as failures because the climber did not survive the descent.*

IV. DISCUSSION

The two-model structure is used to analyze two different questions. The classification model gives an approximation for what predicts success or failure. The regression model on failures gives us the approximation of where on the mountain the failure happens. Together, they show

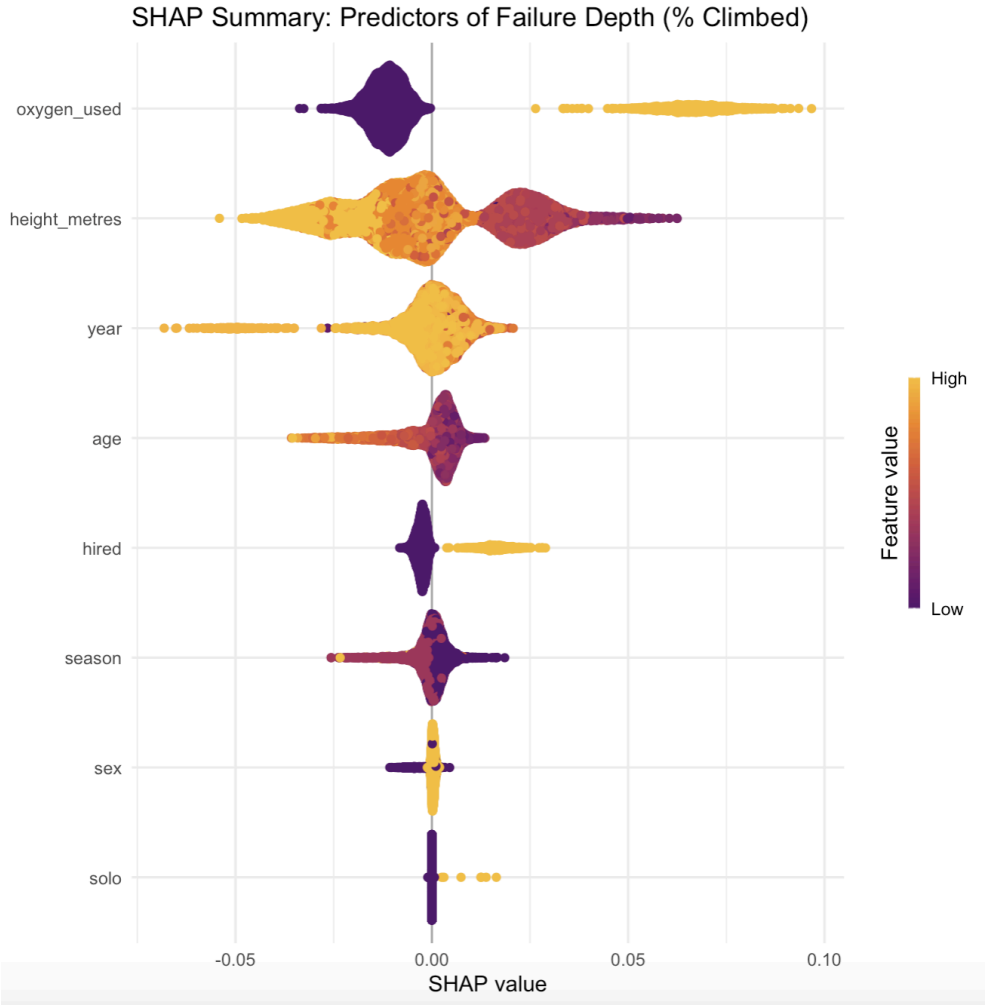
that the dominant failure modes are both early abandonment and resource depletion within reach of the summit. The 38.3% high-uncertainty rate reflects climbers moderate on every predictor in which no single feature dominates. Prior expedition count and summit-window weather are the variables most likely to resolve what the questions ask, perhaps a future exploration if such data were collected. The conformal prediction framework makes this uncertainty explicit, so we don't get a wrong estimation.

The age-hired interaction is the most important finding. Older climbers are less likely to hire guides precisely when the guide effect on predicted success is largest. The model cannot explain why, but the pattern in the data is visible. Temporal leakage is a large limitation. Different climbers from the same expedition can appear across both training and evaluation sets. The 75.8% accuracy measures interpolation over the observed distribution, so it is important to note this model is not for forecasting performance on future expeditions.

The most important take away from this exploration comes from interpretability. Both SHAP and mean decrease in gini both show that using supplemental oxygen and hiring professional assistance are the largest predictors of success on these difficult expeditions. Using this information, we can encourage people that fall within high risk groups (older, climbing during winter, high peak, etc.) to utilize oxygen and hire help.

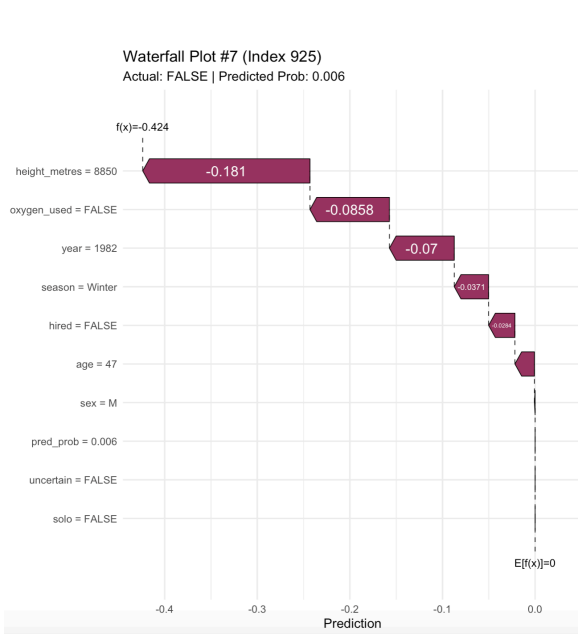
5. Appendix

All Graphs:

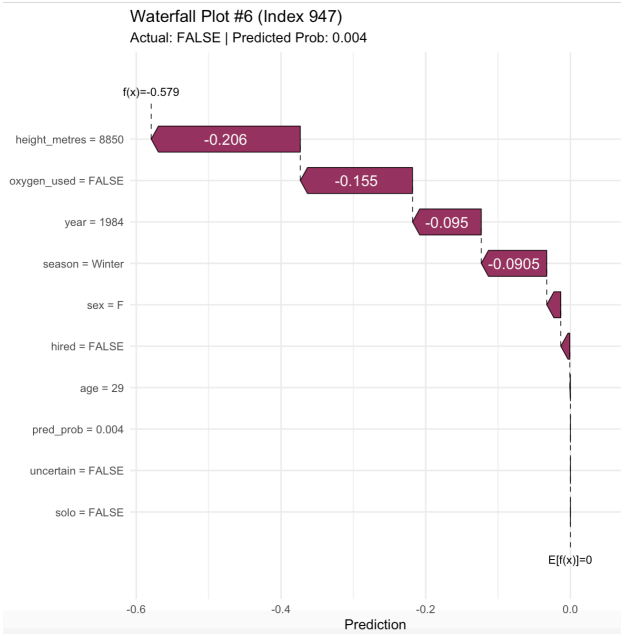


5.1

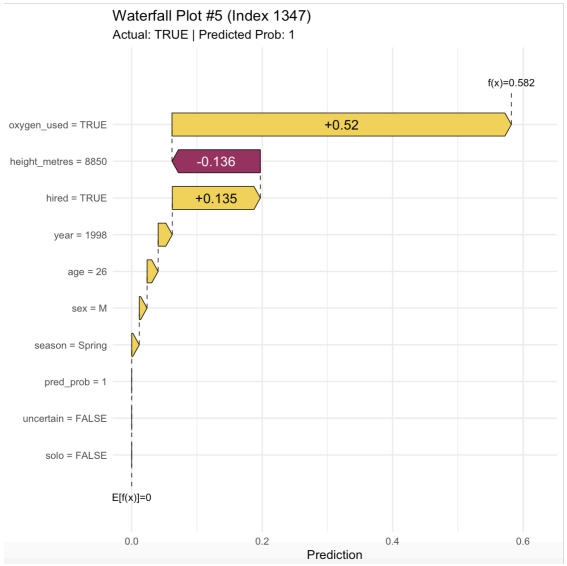
This beeswarm plot displays SHAP values for a random forest regression model predicting the percentage of the mountain climbed before failure among unsuccessful expeditions. Oxygen usage and peak height emerge as the strongest predictors, with oxygen use showing a strong positive impact on failure height (climbers using oxygen tend to fail higher up or even reach the summit). The plot highlights interactions and feature distributions, demonstrating how global interpretability tools like SHAP reveal non-linear relationships and the relative importance of logistical factors (e.g., hiring staff, season) in Himalayan climbing outcomes.



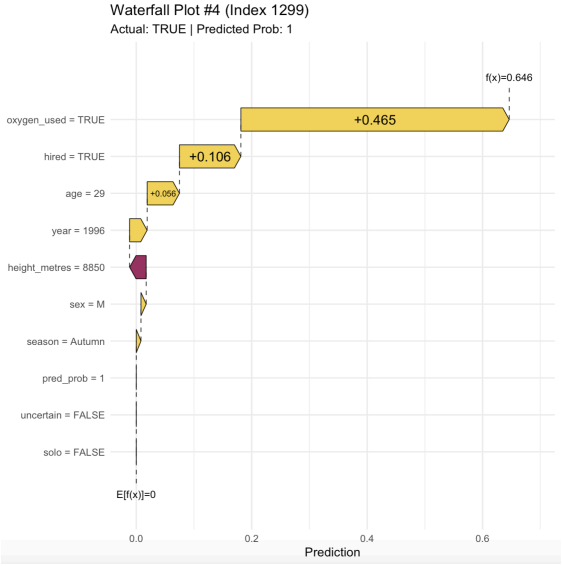
5.3.1



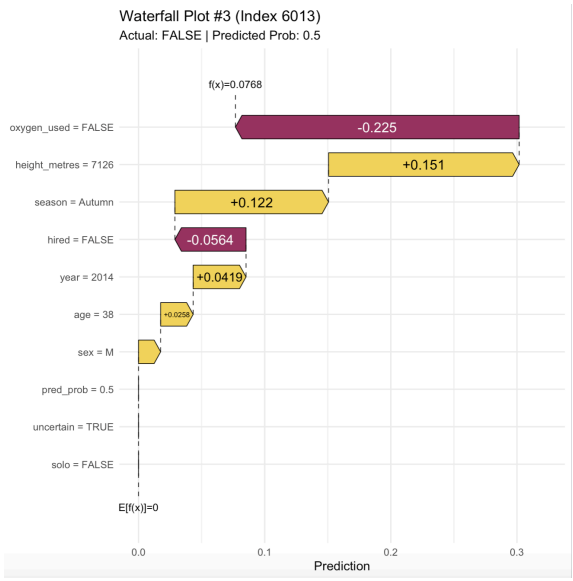
5.3.2

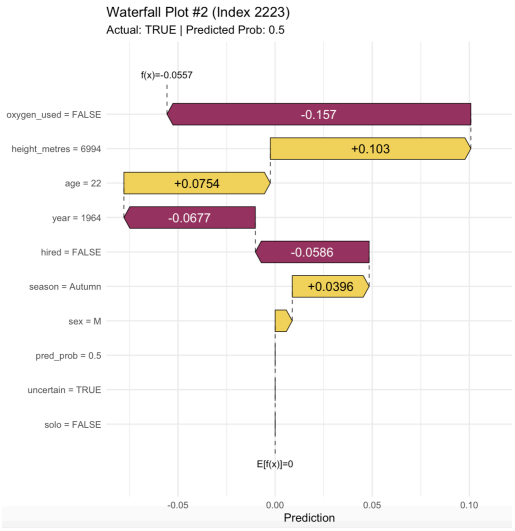


5.3.3



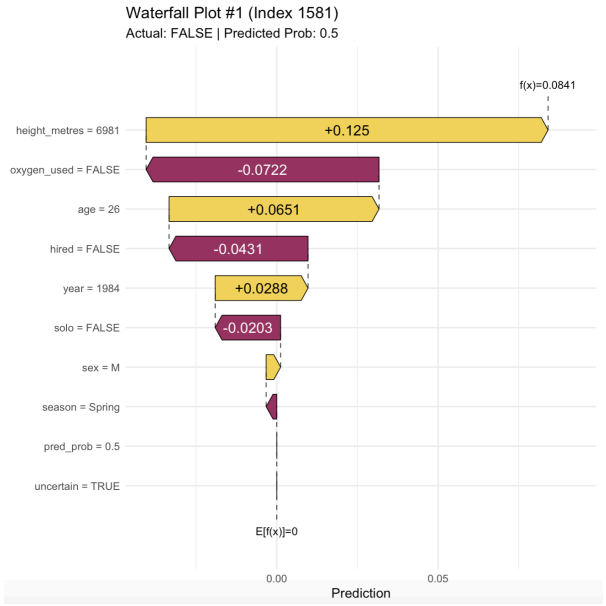
5.3.4





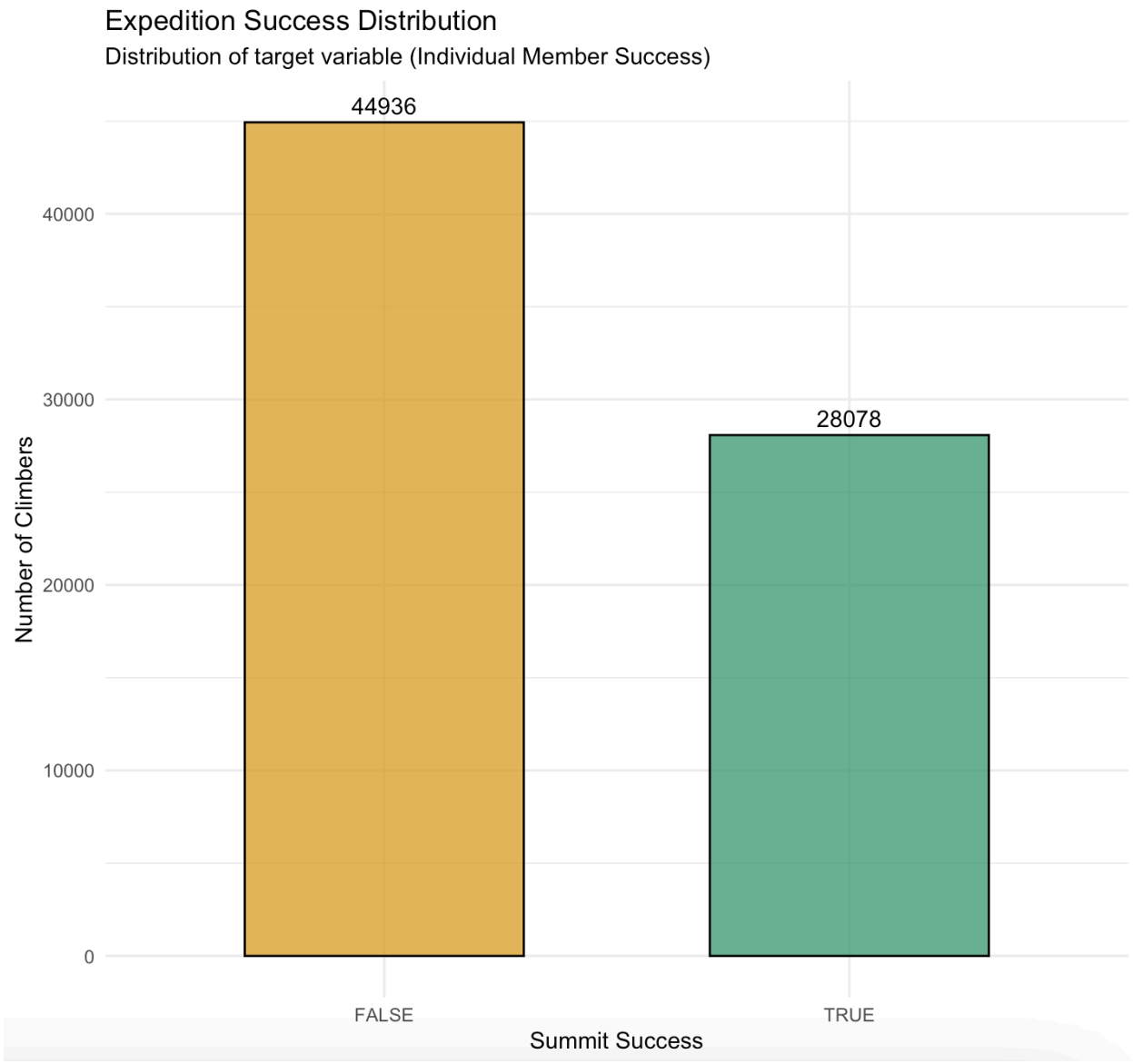
5.3.5

5.3.6



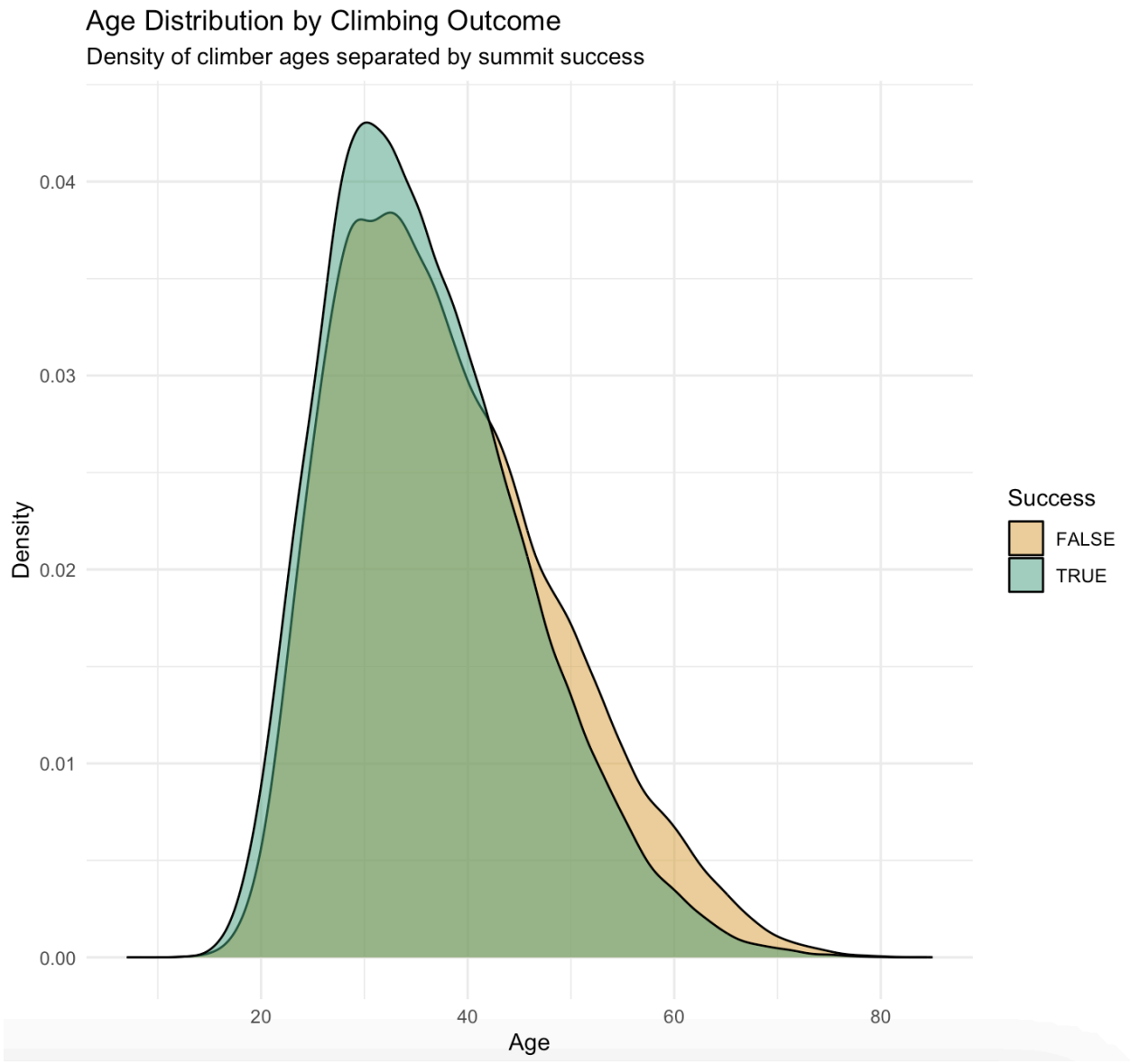
5.3.7

These individual SHAP waterfall plots illustrate how the model arrives at specific predictions for selected climbers, decomposing the output probability into additive feature contributions starting from the base expected value. For high-certainty failures (e.g., Plots #7 and #6 on Everest without oxygen in winter), extreme height and lack of oxygen produce large negative contributions. Successful cases (e.g., Plots #5 and #4) show strong positive pushes from oxygen use and hiring staff. Uncertain cases near 0.5 probability highlight conflicting feature effects and model uncertainty.



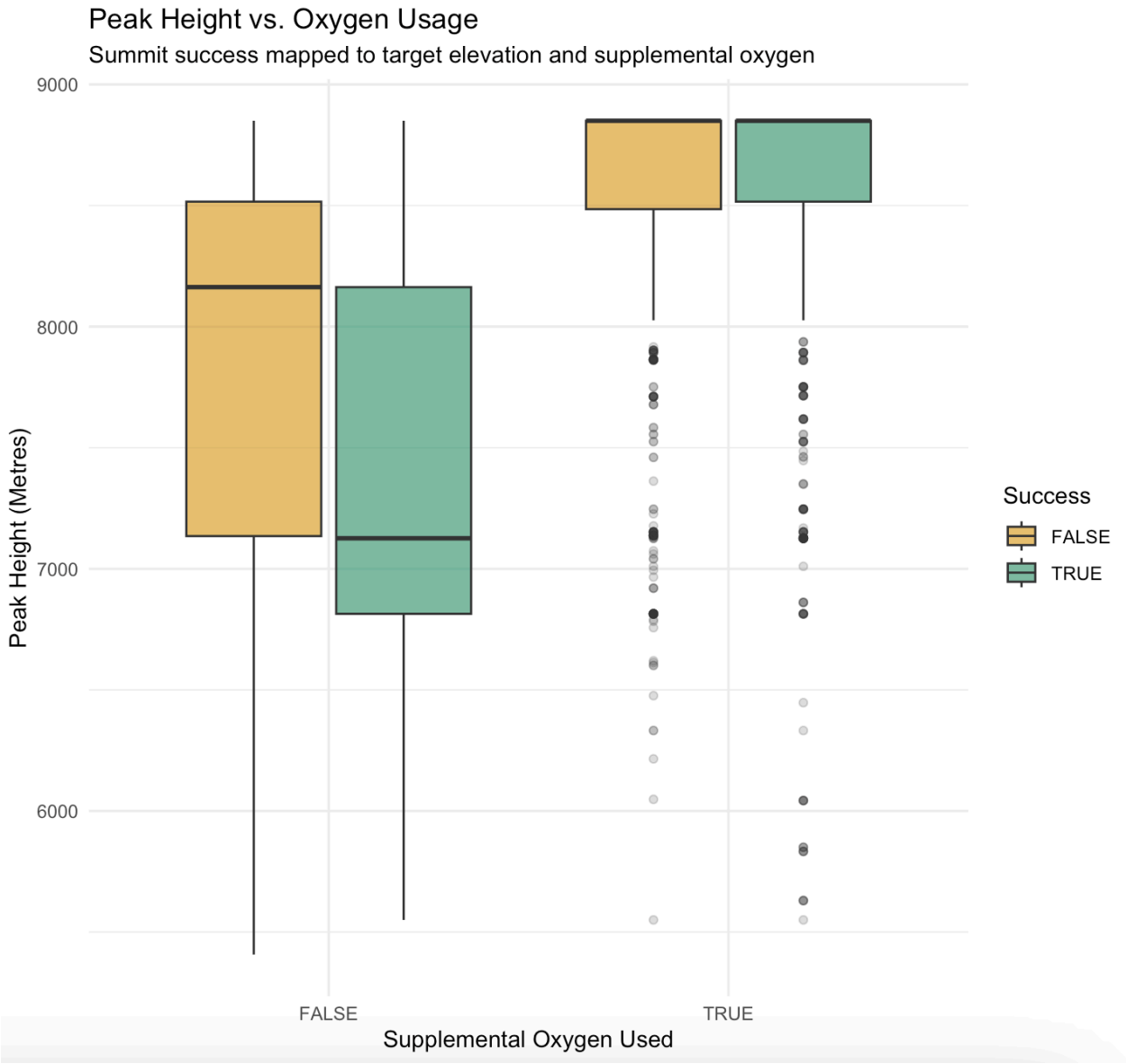
5.8

This bar chart shows the overall class distribution: roughly 38% success rate (28,078 successful vs. 44,936 unsuccessful individual climber outcomes). The imbalance justifies the use of class weights in the random forest and highlights the challenging nature of Himalayan summits.



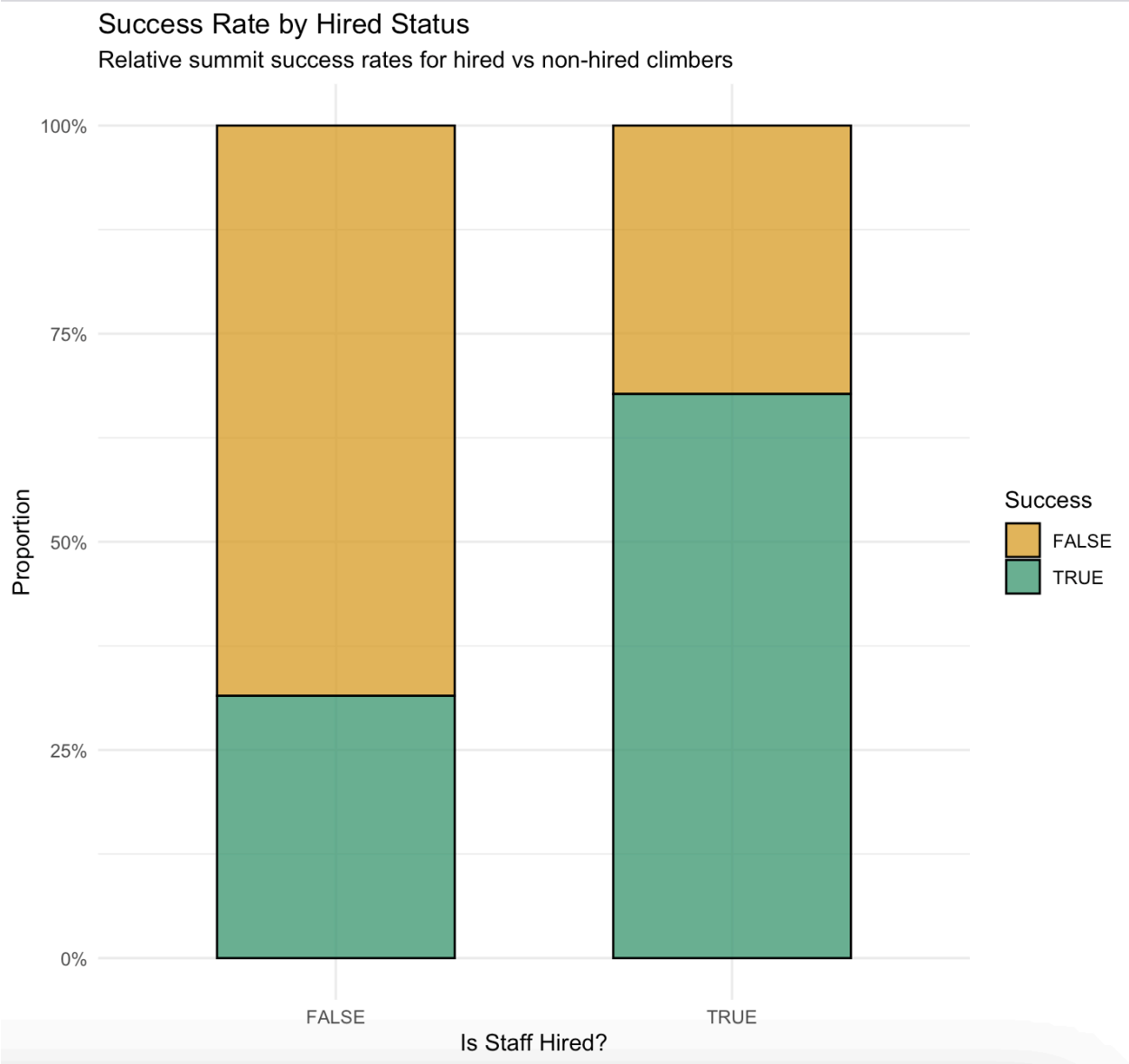
5.9

This overlapping density plot compares age distributions for successful and unsuccessful climbers. Successful climbers tend to be slightly younger on average, with a narrower peak around the late 30s. The visualization reveals a modest but visible age effect, and supports exploratory data analysis before modeling.



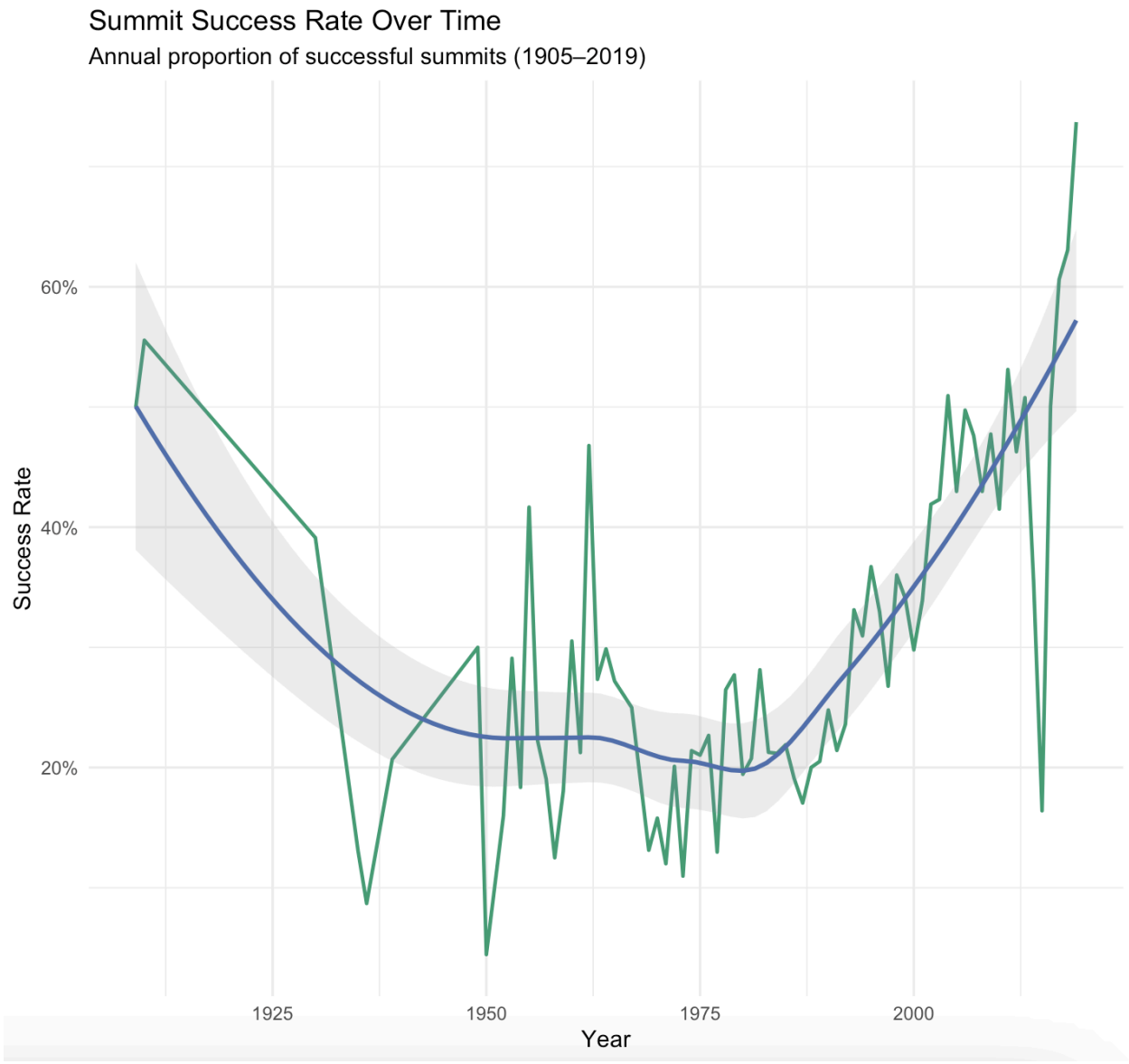
5.10

This boxplot compares peak heights for expeditions with and without supplemental oxygen, colored by success. Oxygen is used more frequently on higher peaks (8000m+), and success rates differ markedly by oxygen use at extreme altitudes. The plot visually confirms the strong interaction between height and oxygen that SHAP dependence plots quantify in greater detail.



5.11

This stacked bar chart compares summit success proportions for climbers with and without hired staff. Hiring staff substantially increases success rate (from ~30% to ~65%), demonstrating the practical value of professional support. This exploratory finding is reinforced by both SHAP and random forest importance analyses.



5.12

This time-series plot shows the annual success rate from 1905–2019, with a LOESS smooth. Success rates declined mid-century but have risen significantly since the 1980s–1990s, likely due to better equipment, weather forecasting, and commercial guiding. This temporal trend is captured by the model (year feature) and provides historical context for modern predictions.

Code:

```
library(randomForest)
library(modeldata)
library(lime)
library(fastshap)
library(shapviz)
library(iml)
library(ggplot2)
library(dplyr)
library(tidyr)
library(scales)
library(caret)

set.seed(123)

expedition_data <- read.csv("Desktop/expeditions.csv")
peak_data       <- read.csv("Desktop/peaks.csv")
member_data     <- read.csv("Desktop/members.csv")

# ── Data Preparation ──────────────────────────────────────────────────────────


merged_data <- member_data %>%
  left_join(peak_data %>% select(peak_id, height_metres), by = "peak_id") %>%
  left_join(expedition_data %>% select(expedition_id, year), by = "expedition_id") %>%
  rename(year = year.y)

df_model <- merged_data %>%
  select(success, age, sex, season, year, oxygen_used, hired, solo, height_metres)

df_model_clean <- df_model %>%
  filter(!is.na(age), !is.na(sex), sex != "Unknown", sex != "") %>%
  mutate(
    success      = as.factor(success),
    sex          = as.factor(sex),
    season       = as.factor(season),
    oxygen_used  = as.factor(oxygen_used),
    hired        = as.factor(hired),
    solo         = as.factor(solo)
  ) %>%
  na.omit()

print(table(df_model_clean$success))
```

```r
# ── Train / Test Split ──────────────────────────────────────────────

train_index <- sample(seq_len(nrow(df_model_clean)), size = 0.8 * nrow(df_model_clean))
train_data  <- df_model_clean[train_index, ]
test_data   <- df_model_clean[-train_index, ]

# ── 5-Fold Stratified Cross-Validation ──────────────────────────────

class_counts  <- table(train_data$success)
class_weights <- max(class_counts) / class_counts

cv_folds <- createFolds(train_data$success, k = 5, returnTrain = TRUE)

cv_accuracies <- sapply(cv_folds, function(idx) {
  fold_train <- train_data[idx, ]
  fold_val   <- train_data[-idx, ]
  fold_model <- randomForest(
    success ~ .,
    data       = fold_train,
    ntree      = 500,
    classwt    = class_weights,
    importance = FALSE
  )
  preds <- predict(fold_model, fold_val)
  mean(preds == fold_val$success)
})

cat("5-Fold CV Accuracy:", round(mean(cv_accuracies), 4),
    "± SD:", round(sd(cv_accuracies), 4), "\n")

# ── Final Model (Balanced Class Weights) ────────────────────────────

rf_model <- randomForest(
  success ~ .,
  data       = train_data,
  ntree      = 500,
  classwt    = class_weights,
  importance = TRUE
)

varImpPlot(rf_model, main = "Standard Random Forest Importance")
```

```r
# ── Exploratory Visualisations ──────────────────────────────────────────────

plot_success <- ggplot(df_model_clean, aes(x = success, fill = success)) +
  geom_bar(alpha = 0.8, color = "black", width = 0.6) +
  scale_fill_manual(values = c("FALSE" = "#E69F00", "TRUE" = "#009E73")) +
  geom_text(stat = "count", aes(label = after_stat(count)), vjust = -0.5) +
  labs(title    = "Expedition Success Distribution",
       subtitle = "Distribution of target variable (Individual Member Success)",
       x = "Summit Success", y = "Number of Climbers") +
  theme_minimal() + theme(legend.position = "none")

plot_season <- df_model_clean %>%
  group_by(season, success) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(season) %>%
  mutate(prop = count / sum(count)) %>%
  filter(success == "TRUE") %>%
  ggplot(aes(x = reorder(season, -prop), y = prop, fill = season)) +
  geom_col(alpha = 0.8, color = "black") +
  scale_y_continuous(labels = percent_format()) +
  labs(title    = "Success Rate by Season",
       subtitle = "Percentage of climbers who successfully summited",
       x = "Season", y = "Success Rate") +
  theme_minimal() + theme(legend.position = "none")

plot_age <- ggplot(df_model_clean, aes(x = age, fill = success)) +
  geom_density(alpha = 0.5) +
  scale_fill_manual(values = c("FALSE" = "#E69F00", "TRUE" = "#009E73")) +
  labs(title    = "Age Distribution by Climbing Outcome",
       subtitle = "Density of climber ages separated by summit success",
       x = "Age", y = "Density", fill = "Success") +
  theme_minimal()

plot_oxygen_height <- ggplot(df_model_clean, aes(x = oxygen_used, y = height_metres, fill =
success)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.2) +
  scale_fill_manual(values = c("FALSE" = "#E69F00", "TRUE" = "#009E73")) +
  labs(title    = "Peak Height vs. Oxygen Usage",
       subtitle = "Summit success mapped to target elevation and supplemental oxygen",
       x = "Supplemental Oxygen Used", y = "Peak Height (Metres)", fill = "Success") +
  theme_minimal()

plot_hired <- ggplot(df_model_clean, aes(x = hired, fill = success)) +
  geom_bar(position = "fill", alpha = 0.8, color = "black", width = 0.6) +
  scale_y_continuous(labels = percent_format()) +
```

```
  scale_fill_manual(values = c("FALSE" = "#E69F00", "TRUE" = "#009E73")) +
  labs(title    = "Success Rate by Hired Status",
       subtitle = "Relative summit success rates for hired vs non-hired climbers",
       x = "Is Staff Hired?", y = "Proportion", fill = "Success") +
  theme_minimal()

plot_year <- df_model_clean %>%
  group_by(year, success) %>%
  summarise(count = n(), .groups = "drop") %>%
  group_by(year) %>%
  mutate(prop = count / sum(count)) %>%
  filter(success == "TRUE") %>%
  ggplot(aes(x = year, y = prop)) +
  geom_line(color = "#009E73", linewidth = 0.8) +
  geom_smooth(method = "loess", se = TRUE, color = "#4C72B0", alpha = 0.2) +
  scale_y_continuous(labels = percent_format()) +
  labs(title    = "Summit Success Rate Over Time",
       subtitle = "Annual proportion of successful summits (1905-2019)",
       x = "Year", y = "Success Rate") +
  theme_minimal()

print(plot_success)
print(plot_season)
print(plot_age)
print(plot_oxygen_height)
print(plot_hired)
print(plot_year)

# ── Global SHAP Values ──────────────────────────────────────────────────

pfun <- function(object, newdata) {
  predict(object, newdata = newdata, type = "prob")[, "TRUE"]
}

X_explain <- test_data %>% select(-success)

shap_values <- fastshap::explain(
  rf_model,
  X           = X_explain,
  pred_wrapper = pfun,
  nsim        = 50,
  adjust      = TRUE
)
```

```r
shp <- shapviz(shap_values, X = X_explain)

plot_shap_beeswarm <- sv_importance(shp, kind = "beeswarm") +
  ggtitle("SHAP Summary: Feature Impact on Summit Success") +
  theme_minimal()

plot_shap_bar <- sv_importance(shp, kind = "bar", fill = "#4C72B0") +
  ggtitle("Global Feature Importance (Mean |SHAP|)") +
  theme_minimal()

plot_shap_height <- sv_dependence(shp, v = "height_metres", color_var = "oxygen_used") +
  ggtitle("SHAP Dependence: Peak Height × Oxygen Use") +
  theme_minimal()

plot_shap_age <- sv_dependence(shp, v = "age", color_var = "hired") +
  ggtitle("SHAP Dependence: Age × Hired Status") +
  theme_minimal()

print(plot_shap_bar)
print(plot_shap_beeswarm)
print(plot_shap_height)
print(plot_shap_age)

# ── Uncertainty Quantification (Conformal Prediction) ─────────────────────────
# Split-conformal approach: calibrate on a held-out calibration set

cal_idx   <- sample(seq_len(nrow(test_data)), size = floor(0.5 * nrow(test_data)))
cal_data  <- test_data[cal_idx, ]
eval_data <- test_data[-cal_idx, ]

cal_probs <- predict(rf_model, cal_data, type = "prob")[, "TRUE"]

cal_labels <- cal_data$success == TRUE

cal_scores <- ifelse(cal_labels,
                     1 - cal_probs,
                     cal_probs)

cal_scores <- cal_scores[!is.na(cal_scores)]

alpha <- 0.10

q_threshold <- quantile(cal_scores,
                        probs = 1 - alpha)
```

```r
eval_probs  <- predict(rf_model, eval_data, type = "prob")[, "TRUE"]
pred_set_includes_true <- (1 - eval_probs) <= q_threshold
pred_set_includes_false <- eval_probs <= q_threshold

coverage <- mean(
  (as.logical(as.character(eval_data$success)) & pred_set_includes_true) |
    (!as.logical(as.character(eval_data$success)) & pred_set_includes_false)
)
cat(sprintf("Conformal coverage at alpha=%.2f: %.4f (target >= %.2f)\n",
            alpha, coverage, 1 - alpha))

# Flag high-uncertainty predictions (both classes included in prediction set)
uncertain_mask <- pred_set_includes_true & pred_set_includes_false
cat(sprintf("High-uncertainty predictions: %d / %d (%.1f%%)\n",
            sum(uncertain_mask), nrow(eval_data),
            100 * mean(uncertain_mask)))

# ── Local Interpretability (SHAP Waterfall Plots) ──────────────────────────


eval_data$uncertain <- uncertain_mask
eval_data$pred_prob <- eval_probs

# Prepare data (same as before)
X_train <- train_data %>% dplyr::select(-success)
X_eval  <- eval_data %>% dplyr::select(-success)

pred_wrapper <- function(object, newdata) {
  predict(object, newdata, type = "prob")[, "TRUE"]
}

# Compute SHAP values (you already have this)
shap_values <- fastshap::explain(
  object = rf_model,
  X = X_eval,
  pred_wrapper = pred_wrapper,
  nsim = 20
)

# Create shapviz object (needs both SHAP values + original features)
sv <- shapviz(shap_values, X = X_eval, pred_wrapper = pred_wrapper)

# Choose interesting cases (your logic)
pred_prob <- pred_wrapper(rf_model, X_eval)
uncertain_mask <- abs(pred_prob - 0.5) < 0.1
uncertain_idx   <- which(uncertain_mask)
```

```r
conf_success_idx <- which(!uncertain_mask & eval_data$success == TRUE)
conf_fail_idx    <- which(!uncertain_mask & eval_data$success == FALSE)

selected_idx <- c(
  head(uncertain_idx[order(abs(pred_prob[uncertain_idx] - 0.5))], 3),
  head(conf_success_idx[order(pred_prob[conf_success_idx], decreasing = TRUE)], 2),
  head(conf_fail_idx[order(pred_prob[conf_fail_idx])], 2)
)

# ==================== WATERFALL PLOTS ====================

# Individual waterfall plots for selected cases
for (i in seq_along(selected_idx)) {
  idx <- selected_idx[i]
  actual <- eval_data$success[idx]
  prob   <- round(pred_prob[idx], 3)

  p <- sv_waterfall(sv, row_id = idx, max_display = 12) +
    labs(title = glue::glue("Waterfall Plot #{i} (Index {idx})"),
         subtitle = glue::glue("Actual: {actual} | Predicted Prob: {prob}")) +
    theme_minimal(base_size = 11)

  print(p)
}

# ── Failure Depth Analysis: Percent Climbed ──────────────────────────

df_failures <- merged_data %>%
  filter(success == FALSE | success == "FALSE") %>%
  filter(!is.na(highpoint_metres), !is.na(height_metres), height_metres > 0) %>%
  mutate(
    percent_climbed = pmin(highpoint_metres / height_metres, 1)
  ) %>%
  select(percent_climbed, age, sex, season, year, oxygen_used, hired, solo, height_metres)

df_failures_clean <- df_failures %>%
  filter(!is.na(age), !is.na(sex), sex != "Unknown", sex != "") %>%
  mutate(
    sex         = as.factor(sex),
    season      = as.factor(season),
    oxygen_used = as.factor(oxygen_used),
    hired       = as.factor(hired),
    solo        = as.factor(solo)
  ) %>%
  na.omit()
```

```r
ggplot(df_failures_clean, aes(x = percent_climbed)) +
  geom_histogram(bins = 30, fill = "#4C72B0", color = "white") +
  scale_x_continuous(labels = percent_format()) +
  labs(title = "Distribution of % Climbed Among Failures",
       x = "Percentage of Mountain Climbed", y = "Count") +
  theme_minimal()

train_index_fail <- sample(seq_len(nrow(df_failures_clean)), size = 0.8 *
nrow(df_failures_clean))
train_fail <- df_failures_clean[train_index_fail, ]
test_fail  <- df_failures_clean[-train_index_fail, ]

rf_fail_model <- randomForest(
  percent_climbed ~ .,
  data       = train_fail,
  ntree      = 500,
  importance = TRUE
)

pfun_reg <- function(object, newdata) predict(object, newdata = newdata)

X_explain_fail <- test_fail %>% select(-percent_climbed)

shap_values_fail <- fastshap::explain(
  rf_fail_model,
  X           = X_explain_fail,
  pred_wrapper = pfun_reg,
  nsim        = 50,
  adjust      = TRUE
)

shp_fail <- shapviz(shap_values_fail, X = X_explain_fail)

plot_fail_beeswarm <- sv_importance(shp_fail, kind = "beeswarm") +
  ggtitle("SHAP Summary: Predictors of Failure Depth (% Climbed)") +
  theme_minimal()

plot_fail_height <- sv_dependence(shp_fail, v = "height_metres", color_var = "oxygen_used")
+
  ggtitle("SHAP Dependence: Peak Height × Oxygen Use (Failures)") +
  theme_minimal()

plot_fail_age <- sv_dependence(shp_fail, v = "age", color_var = "hired") +
  ggtitle("SHAP Dependence: Age × Hired Status (Failures)") +
  theme_minimal()
```

```
print(plot_fail_beeswarm)
print(plot_fail_height)
print(plot_fail_age)
```